

Jones_ProjectSWOTAnalysis

Software Project: Research, Planning, and Design (COS650)

Project: Design and Evaluation of a Predictive System for Bitcoin Volatility Forecasting

Four-quadrant SWOT analysis with direct links to design, scope, evaluation, and proposal planning decisions.

STRENGTHS • The project addresses volatility forecasting rather than simple price direction, which gives it stronger analytical value because volatility is directly tied to risk, uncertainty, and decision support. • Its comparative framework evaluates Historical Volatility, EWMA, Random Forest, and LSTM under one pipeline. That design improves proposal credibility because model complexity is being tested rather than assumed to be better. • The workflow is methodologically disciplined: time-ordered splitting, preprocessing, feature preparation, and RMSE/MAE evaluation. This strengthens internal validity and supports a more defensible research argument. • The system can be framed as a decision-support prototype, not just a prediction engine. That broader framing supports stronger HCI positioning and makes the project easier to connect to responsible use.	WEAKNESSES • The current feature space is dominated by historical OHLCV data. That is useful for baseline forecasting, but it does not fully represent regulatory shocks, sentiment swings, macro events, or derivatives behavior that often influence Bitcoin volatility. • Cryptocurrency markets are non-stationary, so patterns learned in one period may weaken in another. This makes it risky to treat held-out performance as proof of durable forecasting reliability. • The LSTM adds technical depth but also increases sensitivity to hyperparameters, training stability, and limited sample size. In this setting, architectural sophistication can exceed empirical justification. • The project is still stronger as a research prototype than as a deployable system because retraining policy, concept drift response, uncertainty calibration, and maintenance strategy are not yet fully resolved.
OPPORTUNITIES • The strongest technical growth opportunity is feature expansion beyond price history through sentiment signals, blockchain activity, derivatives indicators, or broader macro-financial variables. • The comparative design creates a meaningful research contribution even if simpler baselines perform well. Showing when complexity does or does not add value is academically stronger than forcing a deep-learning success narrative. • There is room to evolve the model into a user-facing analytical tool with dashboards, uncertainty views, and model comparison panels. That would deepen the HCI dimension around trust, clarity, and interpretation. • Because AI-enabled financial analytics remains professionally relevant, the project has value beyond the course as a portfolio piece, proposal foundation, or future research direction.	THREATS • The cryptocurrency environment changes quickly due to regulation, exchange failures, political events, and shifts in investor behavior. Those changes can invalidate learned patterns faster than the model can adapt. • Existing commercial analytics platforms reduce topic novelty. The project therefore has to differentiate itself through methodological transparency, honest claims, and strong benchmarking rather than subject matter alone. • Overfitting is a serious risk in noisy financial data. A model can appear strong on a narrow test window while actually capturing short-lived behavior that does not generalize. • There is also an ethical and practical threat if users interpret the system as financial advice. That makes limitation statements, uncertainty communication, and responsible framing central design requirements.

Strategic Insight and Integration

The strongest strategic position for this project is to present it as a rigorous comparative forecasting and decision-support study, not as a promise that Bitcoin volatility can be predicted with certainty. That framing is more defensible because it aligns the research claims with what the dataset, evaluation design, and model evidence can realistically support.

This SWOT also directly informs design planning. Because dataset scope is a weakness and market instability is a threat, the architecture should remain modular so new data sources, revised features, or updated models can be integrated without redesigning the full system. Because overfitting is a threat, the evaluation plan should prioritize time-aware validation, baseline retention, and careful interpretation of performance gaps. Because misuse is a threat, the interface and reporting strategy should explicitly communicate uncertainty, limitations, and non-advisory intent.

From a project management perspective, the next design priorities are clear: preserve methodological rigor, identify high-value external signals that address current limits, and shape the final system around responsible interpretation rather than raw predictive claims. That direction makes the project stronger academically and more credible during proposal defense because every SWOT category now informs an architecture choice, a validation decision, or a deployment constraint.